# Aniruddha Vijay Savargaonkar

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## RESEARCH INTERESTS

Biomaterials and surface-engineered metallic implants; micro- and nano-scale modification of titanium and alloy surfaces; stem cell-biomaterial interactions and osteogenic differentiation; antibacterial and antifouling biomaterial interfaces; hemocompatibility and blood-material interactions; translational biomaterials for regenerative medicine, blood contacting, and orthopedic applications.

## EDUCATION

**Doctor of Philosophy, Mechanical Engineering**
Colorado State University, Fort Collins | 2021 – 2025
GPA: 3.645
Advisor: Dr. Ketul C. Popat
Dissertation: *Micro/nano scale modification of titanium surface for enhanced mesenchymal stem cell interactions and antibacterial properties.*
Award: Summer Dissertation Fellowship, CSU Graduate School

**Bachelor of Technology, Mechanical Engineering**
Symbiosis International University, Pune | 2016 – 2020
GPA: 3.26

## RESEARCH & ACADEMIC APPOINTMENTS

**Research Scientist**

Colorado State University, Fort Collins | August 2025 – Present

PI: Prof. Matthew J. Kipper

- Lead independent research in biomaterials and nanomaterials with emphasis on bacteria-material and blood-material interactions.
- Develop surface-engineered metallic and polymeric biomaterials and evaluate cytocompatibility, antimicrobial performance, and hemocompatibility using in vitro biological assays.
- Lead manuscript preparation and contribute to competitive research proposal development.
- Mentor graduate and undergraduate researchers through regular meetings, hands-on experimental training, and feedback on data analysis, presentations, and manuscripts.
- Mentor graduate students through mock thesis defense preparation, providing feedback on research framing and technical preparation.
- Contribute to interdisciplinary research planning, and collaborative proposal development across biomaterials and biomedical engineering projects.

**Graduate Research Assistant**

Colorado State University, Fort Collins | August 2021 – August 2025

Advisor: Prof. Ketul C. Popat

- Engineered titanium surfaces using anodization and hydrothermal methods to fabricate nano- and micro-scale features (nanotubes, nanopetals, microporosity) to enhance stem cell interactions while imparting antibacterial properties.
- Evaluated osteogenic differentiation, cytocompatibility, and antibacterial behavior using human mesenchymal and adipose-derived stem cells, bacterial culture models, and quantitative biological assays.
- Assessed hemocompatibility through platelet adhesion, clotting kinetics, and biochemical assays relevant to blood-contacting devices.
- Characterized material surfaces using SEM, XPS, EDS, ICP-MS, fluorescence and immunofluorescence microscopy, and goniometry.
- Contributed to provisional patent applications and presented research at national and international conferences (BMES, SFB, Polish Society for Biomaterials).
- Trained visiting researchers and junior students in laboratory techniques and Biosafety Level-2 (BSL-2) practices.

## TEACHING & MENTORING EXPERIENCE

### Graduate Teaching Assistant

Colorado State University, Fort Collins | August 2021 – May 2025

- Supported 200-level mechanical engineering design courses (MECH 201 & MECH 202) with enrollments of 125-190 students
- Led laboratory sections (~30 students) emphasizing hands-on engineering design and experimental problem-solving.
- Mentored undergraduate students and junior teaching assistants in course content and effective learning strategies.

### Senior Design Project Mentor

Colorado State University, Fort Collins | December 2025 – Present

- Guided a senior design team in the School of Biomedical and Chemical Engineering in experimental planning and technical problem-solving related to biomedical engineering applications.

## TECHNICAL EXPERTISE

### Materials & Surface Engineering

Anodization, hydrothermal synthesis, thin-film deposition, nanostructure fabrication, high-vacuum systems

### Materials Characterization

SEM, XPS, AFM, EDS, ICP-MS, fluorescence microscopy, immunofluorescence microscopy, goniometry

### Biological & Biochemical Assays

Human mesenchymal stem cell culture, adipose-derived stem cell culture, bacterial studies, hemocompatibility assays

**Data & Analysis**

MATLAB, Power BI, JMP, Microsoft Office

## PUBLICATIONS & SCHOLARLY ACTIVITY

**Peer-Reviewed Journal Articles:**
(* denotes equal contribution; **author name bolded**)

1. Abhishek Bhattacharjee, **Aniruddha Vijay Savargaonkar**, Bruno L. Pereira, Paulo Soares, Ketul C. Popat. "Zinc and Strontium-Doped Titania Nanotube Surfaces Enhance Antibacterial Activity and Increase Osteogenic Differentiation of Mesenchymal Stem Cells". *Advanced NanoBiomed Research* (2025)
2. **Aniruddha Vijay Savargaonkar**, Emma Holloway, Ketul C. Popat. "Alkali-Treated, Nanostructured-Micro-Porous Titanium Surfaces Enhance Osteogenic Differentiation of Adipose Derived Stem Cells." *Applied Sciences* (2025) 15, no. 9: 5061
3. Ramesh Singh, Liszt Y.C. Madruga, **Aniruddha Savargaonkar**, Alessandro F. Martins, Matt J. Kipper, Ketul C. Popat. "Tanfloc-Modified Titanium Surfaces: Optimizing Blood Coagulant Activity and Stem Cell Compatibility." *ACS Biomaterials Science & Engineering* (2025)
4. **Aniruddha Vijay Savargaonkar**, Emma Holloway, Liszt Y.C. Madruga, Bruno L. Pereira, Paulo Soares, Ketul C. Popat. "Anti-Bacterial Properties and Hemocompatibility of Alkali Treated Nano-Structured Micro-Porous Titanium Surfaces." *Biomimetics* (2025) 10, no. 2: 115.
5. **Aniruddha Vijay Savargaonkar**, Liszt Coutinho Madruga, Amit H. Munshi, Ketul C. Popat. "Titania Nanotubes Modified with Copper Enhance Osteogenic Differentiation of Adipose Derived Stem Cells." *RSC Advances* (2024) 14, no. 46: 34362-34371
6. Ramesh Singh, Liszt Y. C. Madruga, **Aniruddha Savargaonkar**, Matt J. Kipper, Ketul C. Popat. "Covalent Grafting of Tanfloc on Titania Nanotube Arrays: An approach to Mitigate Bacterial Adhesion and Improve the Antibacterial Efficacy of Titanium Implants." *Advanced Materials Interfaces* (2024)
7. Abhishek Bhattacharjee, **Aniruddha Vijay Savargaonkar**, Muhammad Tahir, Alina Sionkowska, and Ketul C. Popat. "Surface modification strategies for improved Hemocompatibility of polymeric materials: a comprehensive review." *RSC Advances* (2024) 14, no. 11: 7440-7458.*
8. **Aniruddha Vijay Savargaonkar**, Amit H. Munshi, Paulo Soares, and Ketul C. Popat. "Antifouling Behavior of Copper-Modified Titania Nanotube Surfaces." *Journal of Functional Biomaterials* (2023) 14, no. 8: 413.

**Manuscripts in Submission:**

1. **Aniruddha Vijay Savargaonkar**, Ramesh Singh, Ketul C. Popat. "Nano-engineering the Titanium-Tissue Interface: A 15-Year Perspective on Bio-functionalization and Surface Innovation."

**Refereed Chapters in Books:**

**Aniruddha Vijay Savargaonkar**, Ramesh Singh, Ketul C. Popat. "Biofabrication Technologies Other than 3D-Printing &3D-Bioprinting." *Biofabrication in Regenerative Medicine: Basic Technologies, Emerging Modalities, and Applications*. (2025) Springer Nature [Accepted]

**Invited Articles & Professional Publications:**

**Aniruddha Vijay Savargaonkar**. "Micro/Nano Scale Modification of Titanium Surface for Enhanced Biocompatibility." *SFB Biomaterials Forum,* Q4 2025. Invited Featured Article

**Conference Proceedings & Abstracts:**

1. **Aniruddha Vijay Savargaonkar**, Emma Holloway, Liszt Y.C. Madruga, Bruno L. Pereira, Paulo Soares, Alina Sionkowska, Ketul C. Popat. "Anti-Bacterial Properties and Hemocompatibility of Alkali Treated Nano-Structured Micro-Porous Titanium Surface." 34th Polish Society for Biomaterials, Biomaterials in Medicine and Veterinary Medicine Annual Conference (2025), Rytro, Poland.
2. **Aniruddha Vijay Savargaonkar**, Ketul C. Popat. "Enhanced Mesenchymal Stem Cell (MSC) Interaction and Antibacterial Activity on Copper-modified Titania Nanotube Arrays." BMES 2024 Annual Meeting (2024), Baltimore MD.
3. **Aniruddha Vijay Savargaonkar**, Emma M Holloway, Ketul C. Popat. "Antibacterial Behavior of Micro-Porous, Nano-Structured Titanium Surfaces." 2023 Colorado State University Graduate Student Showcase (2023), Fort Collins CO.
4. **Savargaonkar Aniruddha Vijay**, Abhishek Bhattacharjee, Ketul C. Popat. "Antibacterial Activity on Copper Doped Titania Nanotube Arrays." SFB 2023 Annual Meeting & Exposition (2023), San Diego CA.

## LEADERSHIP & PROFESSIONAL SERVICE

**Peer Reviewer (Ad Hoc)**, Springer Nature; Elsevier; MDPI; ACS Publications | 2025 – Present (~15 manuscripts reviewed)

**Abstract Reviewer**, Society for Biomaterials Annual Meeting | 2025, 2026

**Program Chair**, Orthopedic Biomaterials Special Interest Group, Society for Biomaterials | 2023 – 2025

- Coordinated symposium programming and technical session for annual SFB meetings.

**President**, Mechanical Engineering Graduate Association, Colorado State University | 2023 – 2024

**Laboratory Mentor**, Biomaterials Surface Micro/Nano-Engineering Lab, CSU | 2021 – 2025

## PROFESSIONAL MEMBERSHIPS

1. Society for Biomaterials | 2023 – 2025; 2026 – Present
2. Biomedical Engineering Society (BMES)| 2024 – 2025
3. American Heart Association (AHA) | 2025 – Present

## REFERENCES

Available upon request.